

FORET MONTMORENCY, QC, Canada

WMO: 712120

Lat: 47.3228N

Lon: 71.1483W

Elev: 2207

StdP: 13.56

Time Zone: -5.00 (NAE)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-23.5	-18.5	-29.7	1.1	-23.2	-24.6	1.5	-18.1	17.7	10.6	15.9	7.6	3.4	310	0.644

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.5	77.0	63.2	74.2	61.9	71.6	61.0	67.2	72.2	65.3	70.2	63.5	68.6	7.8	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.4	101.8	69.8	63.5	95.2	67.9	61.6	89.0	66.0	33.0	72.4	31.5	69.9	30.1	68.5	73.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.4	12.7	11.4	-31.6	82.1	5.2	2.6	-35.4	84.0	-38.4	85.5	-41.3	87.0	-45.1	88.9		
(5)			-31.8	70.2	5.2	1.6	-35.6	71.3	-38.6	72.3	-41.5	73.2	-45.3	74.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	33.6	5.4	7.8	16.9	31.5	44.3	53.7	59.0	56.7	49.8	38.7	25.5	12.6
	DBStd	20.90	13.42	11.76	11.88	8.71	6.90	6.31	5.23	5.43	7.16	7.88	9.75	11.70
	HDD50	6736	1381	1182	1025	555	205	33	2	9	91	360	734	1159
	HDD65	11469	1846	1602	1490	1004	643	344	197	260	458	817	1184	1624
	CDD50	761	0	0	0	1	27	143	281	217	84	8	0	0
	CDD65	19	0	0	0	0	0	4	10	3	2	0	0	0
	CDH74	259	0	0	0	0	20	56	125	44	14	0	0	0
	CDH80	16	0	0	0	0	1	1	12	1	1	0	0	0
Wind	WSAvg	4.5	4.7	4.9	5.3	5.3	4.5	4.4	4.0	3.8	3.9	4.3	4.5	4.2
Precipitation	PrecAvg	55.7	4.8	3.6	3.7	4.0	4.0	4.7	5.9	4.9	5.1	5.0	5.2	5.0
	PrecMax	64.4	7.8	7.4	5.9	8.1	7.0	7.9	10.6	8.9	10.4	8.7	8.2	7.3
	PrecMin	46.1	1.5	1.8	1.1	1.6	1.7	2.2	1.2	2.3	2.7	1.7	1.6	2.3
	PrecStd	5.7	1.6	1.5	1.2	1.8	1.4	1.6	2.1	1.6	1.6	1.9	1.6	1.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.6	38.7	52.6	62.1	77.1	79.0	81.7	78.4	75.8	66.1	53.8	39.5
		MCWB	42.2	36.3	43.1	48.6	58.6	62.3	65.9	64.7	64.3	56.3	47.9	38.8
	2%	DB	35.4	33.2	42.7	56.1	69.8	75.4	77.4	74.7	70.9	60.7	46.7	35.5
		MCWB	35.0	31.1	34.8	45.7	55.7	61.0	64.0	63.7	62.2	52.9	43.2	35.0
	5%	DB	30.2	30.0	38.4	50.8	64.4	71.9	74.3	71.6	67.5	56.2	42.3	32.7
		MCWB	29.3	28.2	33.4	42.1	53.0	59.4	62.4	61.8	60.6	51.2	40.1	31.8
	10%	DB	25.7	27.2	34.4	45.7	60.1	68.1	71.6	69.1	63.9	52.0	38.4	29.5
		MCWB	24.6	25.5	31.2	38.5	50.7	57.6	61.7	61.0	58.5	48.3	36.7	28.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.3	36.8	45.0	53.3	63.8	68.2	70.0	68.7	67.2	58.9	50.7	39.1
		MCDB	42.7	37.1	50.0	60.3	70.1	73.8	75.4	73.9	71.8	60.7	52.4	39.3
	2%	WB	35.1	31.2	37.3	47.8	59.4	64.8	67.4	66.0	64.5	55.5	44.5	35.0
		MCDB	35.4	32.7	40.3	53.2	65.9	70.0	72.8	70.6	68.7	60.0	45.8	35.4
	5%	WB	29.2	28.2	33.8	42.7	55.4	62.2	65.7	64.2	62.0	51.9	40.4	31.8
		MCDB	29.8	29.7	36.9	48.5	60.5	67.2	70.7	69.3	65.8	54.7	41.6	32.3
	10%	WB	24.5	25.3	31.4	39.5	52.1	59.9	63.7	62.4	59.6	48.7	36.7	28.1
		MCDB	25.4	26.8	34.8	45.0	58.4	65.6	69.0	67.1	63.3	51.7	38.4	28.8
Mean Daily Temperature Range	5% DB	MDBR	21.1	21.2	22.2	19.4	22.5	22.3	21.5	20.9	20.9	15.7	14.9	17.7
		MCDBR	19.1	20.3	26.9	29.1	32.9	29.6	26.7	25.7	26.2	25.0	19.9	17.0
		MCWBR	19.4	18.0	20.5	19.1	19.7	16.4	15.1	15.6	17.5	18.0	16.4	17.9
	5% WB	MCDBR	19.3	18.5	21.1	24.7	25.0	20.6	20.4	19.8	21.5	20.6	17.5	17.2
		MCWBR	19.3	17.1	17.4	18.2	17.9	13.9	13.4	13.3	16.0	16.7	16.1	17.5
Clear-Sky Solar Irradiance	taub	0.244	0.259	0.293	0.321	0.333	0.356	0.371	0.353	0.336	0.311	0.285	0.258	
	taud	2.338	2.288	2.247	2.356	2.468	2.406	2.371	2.427	2.474	2.534	2.493	2.356	
	Ebn at Noon	273	291	295	296	295	287	281	282	278	270	256	251	
	Edh at Noon	22	29	36	36	34	36	37	33	29	23	20	20	
All-Sky Solar Radiation	RadAvg	436	725	1096	1409	1604	1726	1674	1456	1136	668	382	296	
	RadStd	57	88	87	117	168	115	175	95	90	56	43	28	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (33 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page